

Documentation on Transformers

1.. Introduction to Transformers

Transformers are a deep learning architecture introduced in 2017 in the research paper Attention Is All You Need by researchers at [Google Research](https://research.google?utm_source=chatgpt.com).

They are designed to process and understand sequential data such as text, making them highly effective for Natural Language Processing (NLP) tasks like translation, text generation, summarization, and question answering.

Modern AI models such as ChatGPT, BERT, and GPT are built using Transformer architecture.

2.Why were Transformers introduced?

Before Transformers, most NLP systems used:

Recurrent Neural Networks (RNNs)

Long Short-Term Memory Networks (LSTMs)

These models had several limitations:

Problems with RNNs and LSTMs

1. Sequential Processing

Words are processed one by one.

Training becomes slow.

2. Difficulty Handling Long Context

Earlier information may be forgotten when processing long sentences.

3.Limited Parallelization

Cannot efficiently utilize modern GPUs.

Why Transformers Were Introduced

Transformers were developed to:

Process all words simultaneously.

Capture long-range relationships between words.

Improve training speed.

Achieve better performance on language tasks.

Enable large-scale language models.

3. Basic Architecture Overview of Transformers

The Transformer architecture is made up of two main parts:

1. Encoder

Reads and understands the input text.

Converts the input into meaningful numerical representations.

Consists of multiple encoder layers.

2. Decoder

Generates the output text.

Uses information from the encoder to predict the next word.

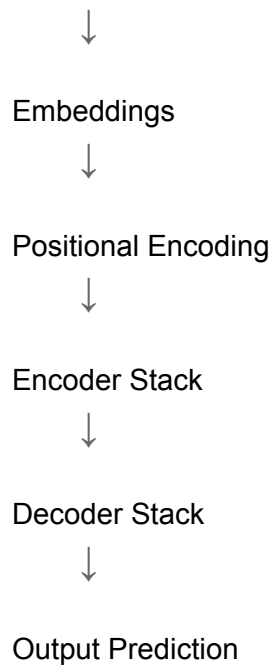
Consists of multiple decoder layers.

Transformer Architecture Flow:

Input Sentence



Tokenization



Main Components

- Input Embeddings

Convert words into numerical vectors that the model can understand

- Positional Encoding

Adds position information to each word since Transformers process all words simultaneously.

- Multi-Head Self-Attention

Helps the model focus on important words and understand relationships between them.

- Feed Forward Neural Network

Processes the information obtained from the attention layer.

- Layer Normalization & Residual Connection

Improve training stability and model performance.

Working Example

Input:
"I love AI"

Process:

I love AI



Embeddings



Positional Encoding



Encoder understands context



Decoder generates output



Final Prediction

4. Role of Attention Mechanism

The Attention Mechanism is the most important component of Transformers.

It allows the model to focus on relevant words while processing a sentence.

Example :

Sentence:

The cat sat on the mat because it was tired.

To understand "it", the model should focus on:

cat

instead of:

mat

Attention helps identify these relationships.

1. Self Attention

Self-attention allows each word to look at every other word in the sentence.

Example:

I love learning AI

When processing "AI", the model pays attention to:

I

love

learning

AI

and determines which words are most relevant.

- Query, Key, and Value

Self-attention uses three vectors:

Query (Q)

Represents the current word asking for information.

Key (K)

Represents information available from other words.

Value (V)

Contains the actual information passed forward.

Attention Score:

$\text{Attention}(Q, K, V)$

=

$\text{Softmax}(QK^T / \sqrt{d})V$

This formula determines how much attention each word receives.

2. Multi Head Attention

Instead of one attention calculation, Transformers use multiple attention heads.

Benefits:

Capture different relationships.

Understand grammar.

Understand meaning.

Improve context understanding.

Example:

One head focuses on:

Subject-Verb relationship

Another head focuses on:

Contextual meaning

Advantages of Transformers:

- ✓ Faster training through parallel processing
- ✓ Better understanding of long text
- ✓ Captures contextual relationships effectively
- ✓ Scalable to billions of parameters
- ✓ Foundation of modern AI models

Applications of Transformers:

1. Machine Translation
- 2 . Chatbots
3. Question Answering Systems
4. Text Summarization
5. Sentiment Analysis
6. Research Paper Assistants
7. Retrieval-Augmented Generation (RAG) Systems
8. Code Generation

Conclusion:

Transformers are a powerful deep learning architecture that revolutionized Natural Language Processing. They were introduced to overcome the limitations of RNNs and LSTMs by using the Attention Mechanism. The architecture consists of encoders and decoders that process text efficiently and understand long-range relationships. Today, Transformers serve as the foundation for advanced AI systems such as GPT, BERT, chatbots, and RAG-based applications.